

JUN (BLAIR) GENG

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EDUCATION

Stanford University

Sep 2023 – Present

PhD in Geomechanics, advised by Prof. Ronaldo Borja

expected December 2026

Dissertation: Implicit and Explicit Smoothed Particle Hydrodynamics for Large Deformation Modeling in Geomechanics

Committee: Paul Segall (chair), Ronaldo Borja, Jon Wren, Nicholas Ouellette, Stephen Monismith

Stanford University

Sep 2020 – Jan 2023

MS in Structural Engineering, GPA 3.9/4.0

Jilin University

Sep 2016 – Jun 2020

BEng in Structural Engineering, GPA 88/100

RESEARCH EXPERIENCE

Stanford University

Sep 2023 – Present

Doctoral Student Researcher

Advisor: Ronaldo I. Borja

- Developed implicit and semi-implicit updated Lagrangian SPH schemes with multiplicative elastoplasticity and Newmark time integration, achieving stable large-deformation simulations at time steps beyond the CFL limit.
- Established a stochastic Smoothed Particle Hydrodynamics framework integrated with Random Field Theory to analyze the probabilistic post-failure behavior of tunnels.
- Proposed a continuum damage model to capture compaction band propagation in porous rocks using the kernel-based approximation and the Smoothed Particle Hydrodynamics method.
- Developed a Smoothed Particle Hydrodynamics framework for undrained loading to study the fault rupture behavior through embankments.

Lawrence Livermore National Laboratory

May 2023 – Aug 2023

Atmospheric, Earth & Energy Graduate Student Intern

Supervisor: Souheil Ezzedine

- Simulated the collapse behavior of a coal mine using the Smoothed Particle Hydrodynamics method.
- Performed verification and validation analysis of the geomechanical response to coal mine collapse.
- Presented a technical report on how geometric configurations and soil conditions affect post-failure behavior.

Stanford University

Sep 2021 – Jan 2023

Graduate Research Assistant

Advisor: Ronaldo I. Borja

- Implemented a variation of the Modified Cam Clay closest point projection return mapping algorithm amenable to the Smoothed Particle Hydrodynamics method.
- Integrated the Drucker-Prager yield criterion into the Smoothed Particle Hydrodynamics method to simulate sand column collapse and landslides.
- Studied the crack propagation in concrete using a gradient-enhanced damage model implemented in MOOSE.

GRANTS & FELLOWSHIPS

Leavell Fellowship on Sustainable Built Environment

2024 – 2025

Stanford University

School of Engineering Graduate Fellowship

2020 – 2021

Stanford University

PEER REVIEWED ARTICLES

- **Geng J**, Borja RI (2026). Implicit and semi-implicit SPH schemes for large deformation problems in geomechanics. *International Journal for Numerical and Analytical Methods in Geomechanics*. <https://doi.org/10.1002/nag.70445>
- del Castillo EM, **Geng J**, Borja RI (2025). A nonlocal kernel-based continuum damage model for compaction band formation in porous sedimentary rock. *Computational Mechanics*, 75(6), 1745-1768. <https://doi.org/10.1007/s00466-024-02540-x>
- del Castillo EM, Fávero Neto AH, **Geng J**, Borja RI (2024). An SPH framework for drained and undrained loading over large deformations. *International Journal for Numerical and Analytical Methods in Geomechanics*, 48(12), 3227–3257. <https://doi.org/10.1002/nag.3790>

MANUSCRIPTS IN PREPARATION

- **Geng J**, del Castillo EM, Borja RI (2026). Stochastic analysis of tunnel face and alcove collapse in coal mine tunnels using smoothed particle hydrodynamics.
- **Geng J**, Yangqing S, Borja RI (2026). Active trapdoor response and soil arching under displacement and stress control.
- del Castillo EM, **Geng J**, Kuhl E, Borja RI (2026). A constitutive neural network-discovered transversely isotropic hyperelastic model for shale rock.

CONFERENCE PUBLICATIONS & PRESENTATIONS

- **Geng J**, del Castillo EM, Borja RI (2026). Probabilistic tunnel collapse mechanism based on smoothed particle hydrodynamics and random field theory. *Geo-Congress 2026 Proceedings*, 208-217. <https://doi.org/10.1061/9780784486719.021>
- del Castillo EM, **Geng J** (2026). Evaluation of retrogressive slope failure in sensitive clays under undrained conditions with Smoothed Particle Hydrodynamics. *Geo-Congress 2026 Proceedings*, 149-158. <https://doi.org/10.1061/9780784486719.015>
- **Geng J** (2025). Stochastic three-dimensional tunnel face stability analysis in spatially variable soils using a meshfree continuum method. Panel presentation at the Engineering Mechanics Institute Conference, Anaheim, CA.
- **Geng J** (2025). Mechanism of coal mine tunnel collapse induced by methane explosion using a meshfree continuum method. Poster presentation at the Blume Earthquake Engineering Center 50th Anniversary Celebration Symposium, Stanford, CA. USA. (Poster).

TEACHING EXPERIENCE

Plasticity Modeling and Computation (CEE 315)

Spring 2024-25

Teaching Assistant, Stanford University

Lecturer: Ronaldo I. Borja

- PhD-level course (~10 students). Held biweekly office hours, and graded the class problem sets. Office hours consisted of first working on additional example problems and then opening the floor to questions.
- Guided students in designing their final projects and implementing different constitutive models at the stress point level based on their research areas.

Mechanics and Finite Elements (CEE 281)

Winter 2023-25

Teaching Assistant, Stanford University

Lecturer: Ronaldo I. Borja

- Graduate-level course (~20 students). Held biweekly office hours, graded the class problem sets, conducted additional classes to help students with their computing assignments, such as implementation of 2D FEM code to solve Poisson's equation over a unit square, as well as more involved 2D BVPs including fluid seepage around a pile, and cantilever beam bending considering different element types in the incompressible limit, and how to avoid volumetric locking.

SERVICE

Peer reviewer

2026 – Present

Acta Geotechnica

Graduate student mentor

2024 – 2025

Stanford University

Supervisor: Morgan Diamond

- Provided support and guidance for the first-year PhD students by meeting quarterly to discuss adjusting to the academic and personal aspects of being a PhD student at Stanford.
- Hosted a large kick-off event in early October and several small mixer events quarterly to foster a strong sense of community and support among the incoming cohort of PhD.

CERTIFICATIONS & LICENSURE

Engineer in Training

Dec 2023

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